

Final Emerging Technology Pitch

Additional process documentation

Project and Technology Overview

This project was a continuation of the AI chatbot and PBL council assignments, which originally used Google Gems and Google Gemini as the basis for the human-AI interaction. At the time, it was not possible to directly embed Gems into other products or sites, so I created separate personas for my original PBL Council project, which was a middle-school interdisciplinary project that challenged groups of students to create a "Golden Record" that represents them and their community, which would be "sent" to space and potentially found by alien civilizations far in the future. I had Google AI Studio build the original bones of a website where the council would be hosted. I then connected a Google AI API to the project and gave the LLM specific global instructions (for audience, safety, tone, style, etc.), and gave more specific prompting to the custom-made personas. I uploaded the project to Github and connected it to Vercel, which I use for web-app project development, and which has cheap or free database hosting. I continued to refine the code (combination of Javascript, HTML, and CSS) and the project using Cursor Pro, which has access to a variety of coding and natural language models, primarily using ChatGPT, Claude, and Cursor's in-house models, Composer and Composer 2. I tested consistently for model performance as well as for UX purposes.

The next phase of the project, which I am submitting here, was to generalize the PBL council creator into its own product for teachers to use to create their own stakeholder councils. This involved creating additional hand-offs between human and AI contributions. I tweaked the format and UI many times to improve clarity, usability, and accessibility. Google Gemini Flash 2.5 and 3.1 are used to support the user's creation of the PBL council, parsing project brief documents and analyzing other user-entered information, such as project goals, learning objectives, and guiding questions. After research AI chatbot safety issues, I consulted with the [KORA benchmark for AI safety](#) and decided to swap Google Gemini as the main chatbot interface with GPT-5.2, which scored much higher on the benchmark for AI child safety than Gemini, and would still be cost-effective during testing and development. Gemini is still used in the Council Creator to parse documents, suggest project details, and find local expert council members.

A defining feature of the project is its implementation of a hybrid approach to human-AI interaction. In the Council Creator, users can enter information in any of the project input

fields, including the project brief, learning objectives, project phases, milestones, and council member bios. Users can then ask the AI to fill in remaining fields with suggestions based on the topic, existing project briefs or other details, the grade level of students, and geographic area. Built into the project is the ability to search for local experts in the topic's discipline or field. After generating AI personas for the council, users can swap out AI council members for human members, either entering information manually (bio, contact info, photo), or by using the LLM to search for a relevant expert. The AI is prompted to search at local universities, community organizations, non-profits, etc. for individuals in a particular field of expertise. If an individual is not found, organizations can be selected with their contact information. For K-12 students this feature is not available on the student-side in the interactive Stakeholder Council (university students and beyond can swap out AI members for human members at any time).

In addition to this 'hybrid' approach in the Council Creator, there are also features in the Stakeholder Council to promote human interaction. When users ask the council members a question or provide a prompt, AI personas are designed to provide structured guidance but not directly give students answers. Instead they answer with reflective prompts, such as "How do you think you could more clearly present the data to an audience?" or "What types of cultural groups might be left out with this approach?" Users can ask one follow-up question to a response. Also, council members will push students to interact with their group members ("Consider talking to your group about...") or resources in their community, with a "Follow-up in your community" section to the response. This feature also uses geographic location taken from a user's IP address (or approximation) to seek out local resources and contact information.

Along with these technical feature, I made several guiding decisions to promote learning design best practices, accessibility and safety, especially for K-12 users, including:

- Increased font size in council responses, using a sans-serif font
- Council responses are limited to a certain number of words/tokens
- Council responses are displayed full-screen (or split-screen when multiple members are prompted)
- Council response text speed is throttled to display slower than a typical LLM, animating word-by-word and pausing after each sentence
- Line breaks are added between paragraphs, bullet points, and questions
- K-12 users are given a character maximum of 144 characters for their prompts (and will have only two uses per visit to the council)

- LLM model shifted to a higher-score model from the KORA AI Child Safety benchmark (revisited often)
- Global prompts given to the LLM for child awareness ("You are speaking with a 6th grader. Explain topics simply when possible.") and blocks of prompting for safety guardrailing ("No harassment; no sexual content; no instructions for weapons; response in a classroom-safe way.." etc.).
- All user inputs and chatbot responses are temporarily recorded on the back-end for a future teacher-specific review/ menu

Learning Design Implications

- The structure of the Council Creator was influenced by [PBLWorks](#), their research, and project examples.
- Suggested project phases, goals, and council roles are designed and prompted in a way to expose users to multiple depth-of-knowledge. (DOK) levels and Bloom's tiers. Some objectives will encourage students to research, synthesize, or gather first-hand resources. Others will encourage students to create artifacts to present to an audience, or engineer multi-disciplinary solutions.
- Responses are "chunked" and limited to prevent cognitive overload, to meet students at a developmentally appropriate level, and to provide achievable goals to promote self-efficacy and engagement.
- Metacognitive skills are exercised through reflective prompts for teachers and students. The Council Creator generates a pre-Launch set of reflective questions for teachers to examine biases, consider students' experience, and incorporate a wider community-level context. Future features may include student self-assessment rubrics and reflections on their responsible use of AI.

Feasibility and Constraints

The project is easily accessible to teachers on the internet. It will not be publicly available for use (as of now), as additional stress-testing and UX feedback is needed, and my personal APIs are connected to the projected. So a related constraint is the cost of the tool's use, as it utilizes LLMs at various stages. One option is to make the project publicly available on Github for users to connect their own APIs to, or for an organization, such as a school or district to use it with an organizational AI subscription.

As of this writing, the tool has not been tested for mobile or tablet use, which could be a constraint for some audiences.

For K-12 audiences (I have options for Grade 3 and older currently), there are a number of considerations and concerns with students using AI tools such as chatbots. While I have attempted to structure the project to keep this audience in mind, it may not be feasible, or desirable to use the tool directly with younger students. A recommended path would be Grade 3-8 students to use with close guidance from an adult, such as students submitting their questions to the teacher, who then uses the tool. High School students would still need guidance and monitoring for the tool's appropriate use.

Another feasibility consideration is the tool's practical use and learning curve for users, teachers and students alike. Mitigating this could involve significant UX testing and engineering, and revamping the UI for more guided use. I have begun exploring options (linked in my Figma presentation) for UI changes that are specific to different ages and education levels.

The main question for users will be if this is a tool that is *additive* to their experience as a teacher or student. I believe it presents users with opportunities they could not access otherwise, and provides a structured, safe method for interactive with professionals and experts in a field.

Learning Reflection

There are two main strands of learning I experienced over the course of this project. The first is the technical knowledge that I acquired to launch such a project from idea to shipping a live, working product. I had some familiarity with aspects of how to achieve the task, but was able to incorporate learning from this course into existing technical skills I have practice over the last several years. Although I had used chatbots before, I had not prompted and built my own, or incorporated one into an existing or new project. Having to consider technical, pedagogical, moral/ethical, and philosophical issues during the project was a challenge, but worthwhile endeavor. I often reflected on design decisions and considered their efficacy, practicality, and pedagogical implications. There was not, and still is not, a clear answers on some of the thornier issues surrounding AI use, especially for younger audiences. I consider this aspect of the project to be a major takeaway and opportunity for future learning, especially as (A) innovative technology continues to drive forward and bring with it fraught societal issues and (B) public sentiment sways as the pressures from such technology proliferate. This will guide some of my thinking as I continue to develop learning products that use new technologies, from a personal, professional, and business standpoint. The answers may not ever be completely clear, but I look forward to continuing this learning process as I work toward creating memorable and effective experiences for learners!

Suggested Council Creator use (use second link at top of this document)

1. Upload an existing PBL project brief, such as one found at PBLWorks or any other document.
2. Review the suggested Essential Question and Learning Objectives. Edit, refine, and reorder.
3. Ask for Suggested project phases. Consider writing 1-2 phases manually and asking the AI for the rest.
4. Use Generate Council. Try the Regenerate button on one member and ask for a local expert for one member. Select which phases each member should be available for.
5. Click the Teacher Pre-Launch Reflection check box (Step 7) and review the reflective prompt.
6. Launch the Council!
7. Once in the council, try to replace one of the AI members with a human member (manually or with Search).
8. Click through the project phases and try using the prompt input to ask 1-2 council members a question at one time.
9. Check if local community resources are relevant to the project and your question.
10. Your council should be saved from the main site menu now, and you can even share the link with others (note: please don't share publicly, see Note below).

Note: since this project is connected to my personal APIs, please limit prompting to 3-4 uses as multiple people may be reviewing this project at once. Thank you!